RIVER VALLEY HIGH SCHOOL
YEAR 6
PRELIMINARY EXAMINATION II

H2 BIOLOGY 9648

PAPER 1
28 SEP 2015

1 HOUR 15 MIN

NAME ____

CLASS 6 (   )

INDEX NO. ____

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, class and index number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

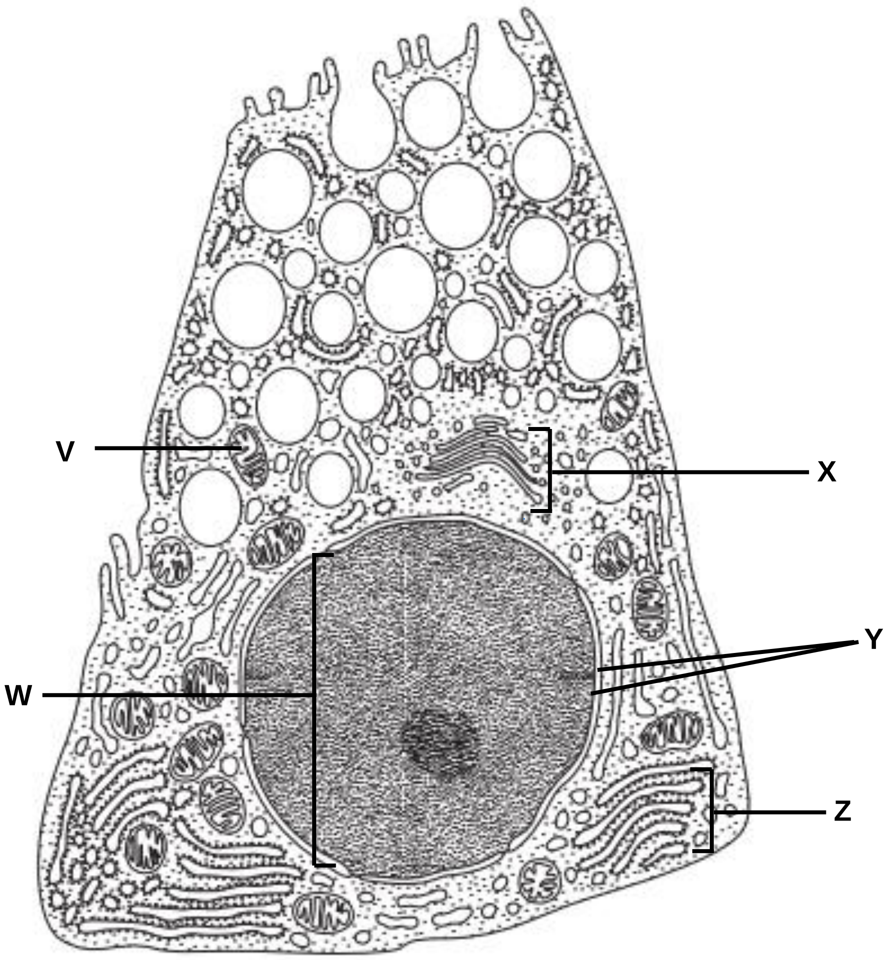
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

This Question Paper consists of **25** printed pages.

For each question, there are four possible answers, **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

1. The diagram shows a drawing of a gut cell.



Which of the following statement(s) is correct?

1. V is the site of ATP synthesis during aerobic respiration.
2. X and Z are both involved in the synthesis of lipid and steroid.
3. W and V both contain template for transcription.
4. Y is continuous with the membrane of Z and X.
5. Structures V, W, X, Y and Z are found in animal cells but not in plant cells.

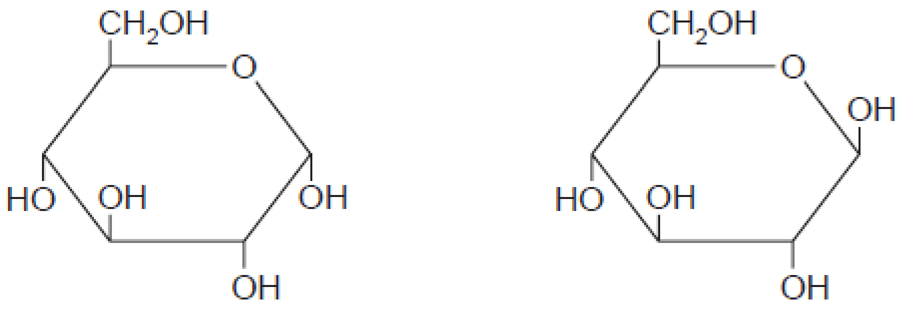
A 1 and 3 only

B 1, 2 and 4 only

C 1, 3 and 4 only

D All of the above

2. The diagram shows the structure of two molecules X and Y.



molecule X   molecule Y

Which statements are correct?

1. Many of molecule X join to form amylose.
2. Many of molecule Y join to form cellulose.
3. Many of molecule X join to form amylopectin.
4. Many of molecule Y join to form glycogen.

A 1 and 2 only

B 1 and 3 only

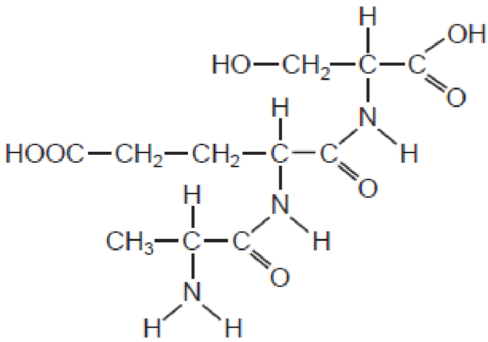
C 2 and 3 only

D 1, 2 and 3 only

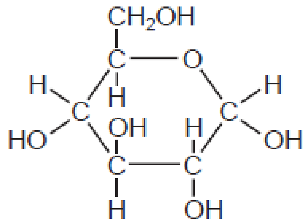
3. Samples of a mixture of biological molecules were tested using Benedict's reagent, biuret solution and ethanol. After testing, the solutions were blue with Benedict's reagent, purple with biuret solution and cloudy with ethanol emulsion test.

Which molecules could the mixture contain?

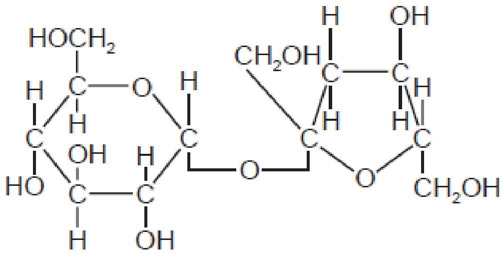
W



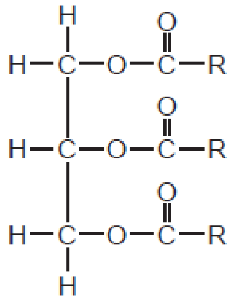
X



Y



Z



A W, X and Y

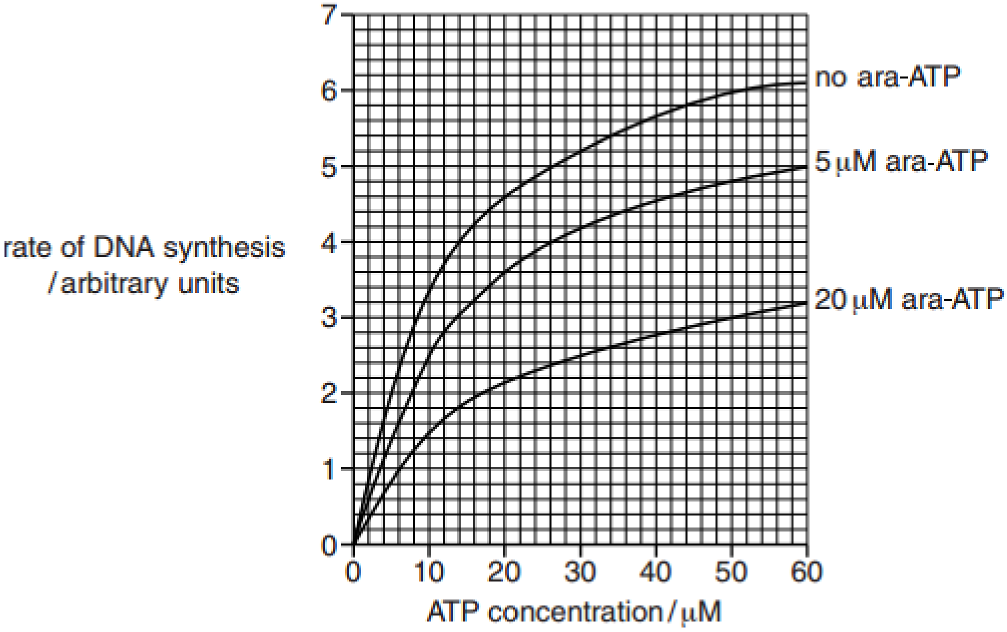
B W, X and Z

C W, Y and Z

D X, Y and Z

4. DNA polymerase requires ATP to perform its catalytic activity. A chemical, ara-ATP, affects DNA polymerase activity.

The graph below shows the effect of different concentrations of ara-ATP on the rate of DNA synthesis.



Using the information from the graph, which is a valid conclusion(s) about ara-ATP?

1. ara-ATP acts as a competitive inhibitor of DNA polymerase.
2. Binding of ara-ATP resulted in a change of active site shape of DNA polymerase.
3. Increase in ATP concentration overcomes the effect of ara-ATP on DNA polymerase.
4. An increase ara-ATP concentration results in a decrease the amount of DNA molecules synthesised in a given time.

A 1 only

B 4 only

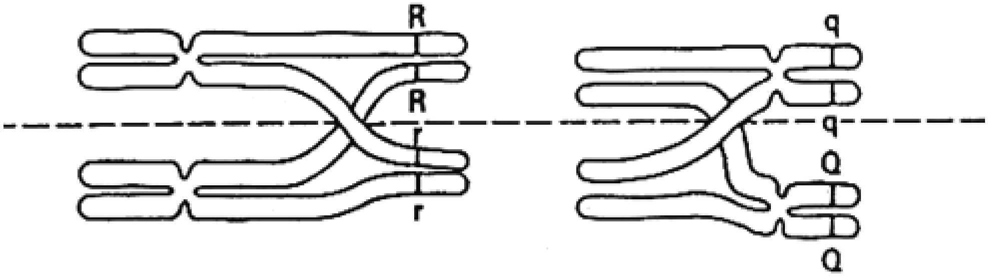
C 1 and 3 only

D 2 and 4 only

5. What are the conditions in a human cell just before the cell enters prophase?

	number of chromatids	number of molecules of DNA in nucleus	spindle present	nuclear envelope present
A	46	46	yes	no
B	92	46	no	yes
C	46	92	yes	yes
D	92	92	no	yes

6. The figure below shows the arrangement of two pairs of homologous chromosomes at metaphase of the first meiotic division. R and r, Q and q are two allelic pairs.



What will be the genotypes of the gametes subsequently produced?

A R q r Q

B Rr qq Rr QQ

C Rq rQ Rq rQ

D Rq rq RQ rQ

7. For organisms undergoing sexual reproduction, a reduction division occurs before fertilisation.

Which reasons explain why this is necessary?

1. increase genetic variation
2. prevent doubling of the chromosome number
3. reduce the chances of mutation

A 1 only

B 2 only

C 2 and 3 only

D 1, 2 and 3

8. During semi-conservative replication of DNA in eukaryotic cells, the following processes occur.

1. Free nucleotides are hydrogen bonded to those on the exposed strand.
2. Hydrogen bonds are broken between the complementary base pairs.
3. The cell receives the signal to begin to divide.
4. Covalent bonds form between adjacent nucleotides on the same strand.
5. The DNA double helix is unwound.

Which shows the correct order of some of the processes?

A 3 → 1 → 2 → 4

B 3 → 2 → 4 → 5

C 5 → 2 → 1 → 4

D 5 → 2 → 3 → 1

9. In most organisms, all possible triplets of bases in DNA code for an amino acid, except for the coding strand ‘stop’ triplets ATT, ATC and ACT. The coding strand is complementary to mRNA.

In the prokaryote, *Methanosarcina barkeri*, the coding strand triplet ATC codes for an unusual amino acid, pyrrolysine.

The non-coding strand of a section of DNA in *M. barkeri* has the following sequence of bases.

TTTTTATTGTATTACTAGTGTTAATGA

How many amino acids can be made into a peptide by *M. barkeri* from the coding strand of this region of DNA.

A 5

B 6

C 7

D 9

10. The table gives the tRNA anticodons for five amino acids.

Amino acid	Anticodon (tRNA)
asparagine	UUA
methionine	UAC
proline	GGA
threonine	UGG
lysine	UUU

A cell makes a polypeptide with the first five amino acids shown below:

Position	1	2	3	4	5
Amino acid	methionine	asparagine	threonine	proline	asparagine

As a result of gene mutation, the following polypeptide is synthesised instead:

Position	1	2	3	4	5
Amino acid	methionine	lysine	proline	proline	asparagine

What could account for the difference in amino acid sequence of the two polypeptide chains?

A A deletion in the DNA code at position 2 and an insertion in the DNA code at position 3.

B An insertion into the DNA codes at position 2 and a deletion in the DNA code at position 3.

C A change in the sequences of the second and third nucleotides at position 2 and 3.

D A single deletion in the DNA code at position 2.

11. The tobacco mosaic virus has RNA rather than DNA as its genetic material. In a hypothetical situation, RNA from a tobacco mosaic virus is mixed with proteins from a related DNA virus Z, resulting in a hybrid virus. If this hybrid virus were to infect a cell and reproduce, what would the resulting "offspring" viruses be like?

A Tobacco mosaic virus

B Virus Z

C A hybrid virus with tobacco mosaic virus RNA and virus Z protein

D A hybrid virus with virus Z DNA and tobacco mosaic virus protein

12. Tamiflu is an antiviral drug that is used in the treatment of influenza. It acts as an inhibitor of the neuraminidase on the influenza virus.

Which stage of the influenza life cycle will be inhibited by Tamiflu?

A Attachment

B Replication of viral genome

C Assembly

D Release

13. Which correctly describe the transfer of DNA in prokaryotes?

	Binary fission	Transduction	Transformation
A	DNA transferred by bacteriophage from one bacterium to another	Bacterial cell takes up foreign DNA from culture medium	Single strand of F plasmid transferred from one bacterium to another
B	DNA transferred by bacteriophage from one bacterium to another	Single strand of F plasmid transferred from one bacterium to another	Bacterial cell takes up foreign DNA from culture medium
C	Bacterial chromosome and plasmids pass to daughter cells	DNA transferred by bacteriophage from one bacterium to another	Single strand of F plasmid transferred from one bacterium to another
D	Bacterial chromosome and plasmids pass to daughter cells	DNA transferred by bacteriophage from one bacterium to another	Bacterial cell takes up foreign DNA from culture medium

14. A single mutation has occurred in an *Escherichia coli*. Glucose and lactose are both absent in the culture medium where the mutant *E. coli* is grown in. An analysis of proteins synthesised by mutant *E. coli* found substantial amount of β-galactosidase, transacetylase and permease.

Which of the following mutations could have taken place in the *E. coli* cell?

1. Mutation in the operator of the *lac* operon
2. Mutation in the *lacI* gene
3. Mutation in the promoter of the *lac* structural genes

A 1 only

B 2 only

C 1 and 2 only

D 2 and 3 only

15. Which of the following statement(s) is true of the end-replication problem of linear DNA?

1. When a linear DNA molecule replicates, a gap is left at the 3' end of each new strand because DNA polymerase can only add nucleotides to a 5' end.
2. Telomerase prevents the end-replication problem from occurring.
3. Repeated rounds of replication produce shorter and shorter DNA molecules.
4. Prokaryotes do not have the end-replication problem.

A 1 and 2

B 1 and 4

C 2 and 3

D 3 and 4

16. Which methods of regulation of gene expression in eukaryotes take place prior to translation?

1. Proteolytic cleavage of signal peptides from precursor protein
2. Alteration of chromatin conformation by DNA methylation
3. Alternative splicing of pre-mRNA
4. Addition of lysine to histone tails

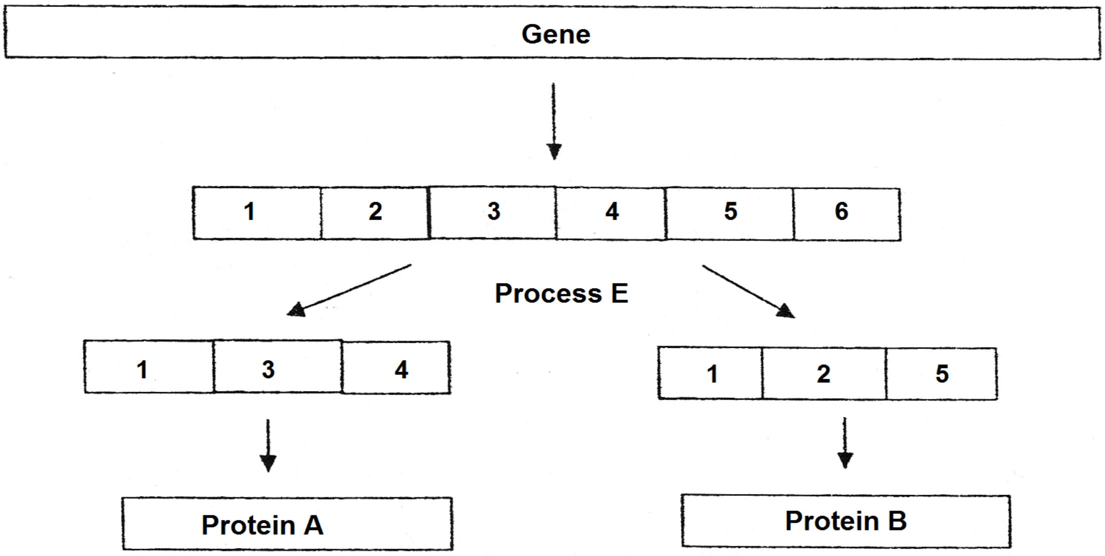
A 1 and 2 only

B 2 and 4 only

C 1, 2 and 3 only

D 2, 3 and 4 only

17. The diagram shows transcription and processing of pre-mRNA into two different mature mRNAs, leading to the production of two different proteins.



Which of the following statement explains the significance of process **E** to eukaryotic cells?

A This is a way of saving space in a small genome.

B This is the process where introns are edited out of the RNA.

C This is the process for a single gene to code for structurally variant versions of a protein.

D This is the way to remove untranslated regions.

18. The human papillomavirus may transform its target cell into a cervical cancer cell, which can lead to a tumour.

Which statements correctly explain this transformation?

1. Loss of function mutation of the gene that codes for protein that causes cell death.
2. Amplification of the gene that codes for proteins that stimulate normal cell division.
3. Gain of function of mutation of gene that code for protein that inhibits cell cycle.
4. Translocation of gene that code for protein that repair damage DNA to strong regulatory sequence.

A 1 and 2 only

B 2 and 4 only

C 1 and 4 only

D 3 and 4 only

19. A plant is known to be heterozygous at two gene loci, X and Y. The pollen grains from this plant are used to fertilise another plant of the same genotype. What is the probability that an embryo will be homozygous dominant at one locus.

A 1 in 4

B 3 in 8

C 5 in 8

D 1 in 16

20. A couple underwent a pre-marital genetic screening. Their genetic profile for two selected characteristics is shown in the table below.

	Xiong Xiong	Ling Ling
Red-green colour blindness	Negative	Negative
Rhesus factor	Negative	Positive

A detailed report states that Ling Ling is heterozygous at both loci.

What is the probability that their first child will be a rhesus negative, red-green colour blind boy?

A 0

B 0.0625

C 0.125

D 0.25

21. Pollen from an aubergine plant with purple flower and purple fruit was transferred to a pure-bred aubergine plant with white flower and white fruit. Seeds were collected and grown, giving 350 plants with the following phenotypes.

purple flowers and purple fruit	153
purple flowers and white fruit	29
white flowers and purple fruit	21
white flowers and white fruit	147

Which of the following statements best explain the results?

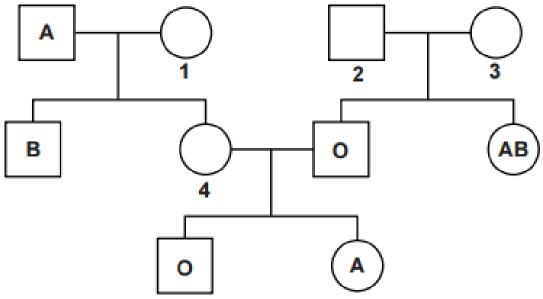
A The gene controlling colour of flower is epistatic over the gene controlling fruit colour.

B The gene controlling colour of flower and fruit colour are found far apart on the same chromosome.

C The gene controlling colour of flower and fruit colour are found adjacent to each other on the same chromosome.

D The gene controlling colour of flower and fruit colour assort independently during meiosis.

22. The pedigree diagram below shows the inheritance of the ABO blood group system. The blood group of some of the individuals is given in the pedigree.



Which of the following correctly identifies and genotype of individuals 1, 2, 3 and 4?

	1	2	3	4
A	I^A	I^O	I^B	I^B
B	I^BI^O	I^OI^O	I^AI^B	I^AI^A
C	I^AI^O	I^OI^O	I^BI^O	I^BI^O
D	I^BI^O	I^BI^O	I^AI^O	I^AI^O

23. A black-haired female hamster was crossed with a white-haired male hamster. Nine offspring were born. Two were white-haired males, three were white-haired females and all the others were black-haired females.

Several statements about the inheritance of hair colour in hamsters are listed below:

1. Hair colour is a sex-linked trait.
2. Hair colour is an autosomal trait.
3. The allele for black hair may be dominant to the allele for white hair.
4. The allele for black hair may be recessive to the allele for white hair.

From the evidence above, which of the statements can be correctly concluded about the inheritance of hair colour in hamsters?

A 1 and 3 only

B 1 and 4 only

C 2 and 3 only

D 2, 3 and 4 only

24. Two pure-bred lines of different plant varieties, which differed markedly in height, were crossed. The height of the two parental varieties and their offspring were measured to the nearest cm. The number of plants in each height category was counted.

The table below shows the results.

Height of plant / cm		4-6	7-8	9-10	11-12	13-14	14-15	15-16	16-17	17-18
number of plants	Parental	5	375	177				352	955	10
	Offspring			13	544	974	48	2		

What is the cause of the phenotypic variation shown in height of plants within the two parent varieties and their offspring.

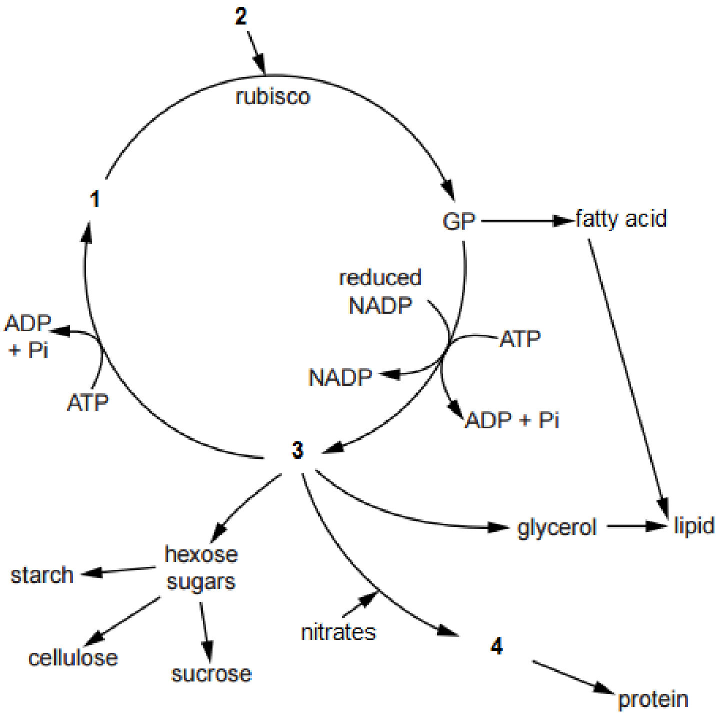
A Linkage and crossing-over at meiosis

B Segregation and independent assortment of alleles

C Various environmental factor

D Additive effect of different genes

25. The diagram shows some of the reactions that take place inside a mesophyll cell.



Which of the following correctly identifies 1, 2, 3 and 4?

	1	2	3	4
A	Glyceraldehyde-3-phosphate	Pyruvate	Ribulose bisphosphate	Amino acid
B	Amino acid	Ribulose bisphosphate	Glyceraldehyde-3-phosphate	Oxaloacetate
C	Ribulose bisphosphate	Carbon dioxide	Glyceraldehyde-3-phosphate	Amino acid
D	Oxaloacetate	Pyruvate	Glyceraldehyde-3-phosphate	Carbon dioxide

26. Antimycin A and oligomycin are metabolic poisons which affect cellular respiration in eukaryotes. A description of how these poisons affect cellular respiration is shown in the table below.

Antimycin A	Binds to electron carrier, cytochrome *b*, cytochrome *b* cannot donate electrons to other electron carriers.
Oligomycin	Binds to Fo subunit of ATP synthase, blocking proton channel.

Which of the following statement is incorrect?

A Poisoning by antimycin A and oligomycin will result in decrease in ATP yield.

B Poisoning by antimycin A and oligomycin will decrease the amount of pyruvate produced.

C Oxygen is used in cell poisoned by oligomycin but not in cell poisoned by antimycin A.

D Cellular levels of reduced NAD increases with antimycin A poisoning but not with oligomycin poisoning.

27. In plants adapted to cold conditions, their cell surface membranes change as the weather gets colder, allowing the plants to carry out exocytosis.

Which change(s) occurs in their cell surface membranes?

1. An increase in the ratio of proteins to unsaturated phospholipids
2. An increase in the ratio of unsaturated phospholipids to saturated phospholipids
3. A decrease in the ratio of unsaturated phospholipids to saturated phospholipids
4. An increase in the proportion of cholesterol in the membrane

A 1 only

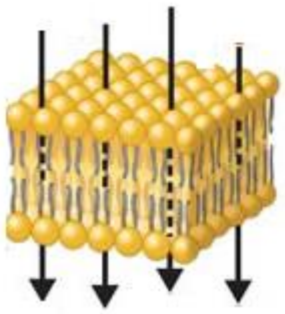
B 4 only

C 2 and 3 only

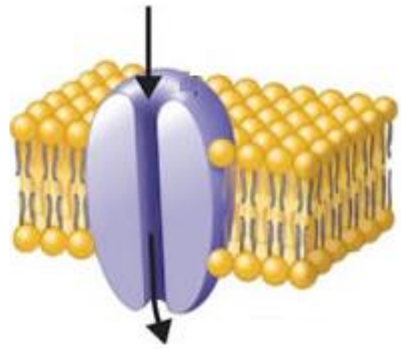
D 2 and 4 only

28. The transport of four different molecules across the cell membrane is shown below. Which of the following illustrates how the ligand of testosterone receptor enters the cell?

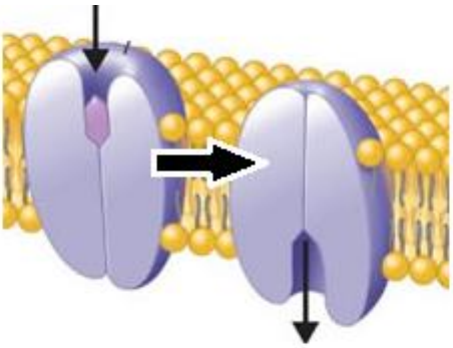
A



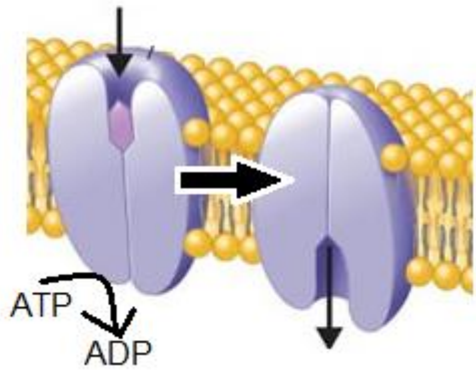
B



C



D

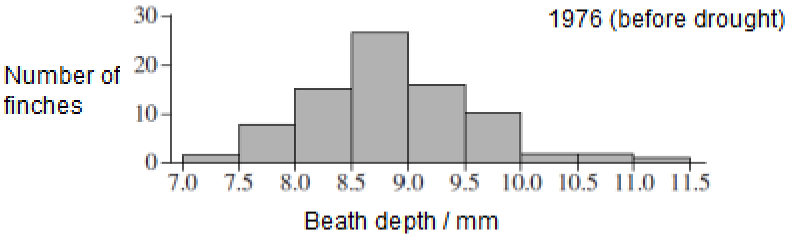


29. The ground finch, *Geospiza fortis*, is a species of bird which lives on a small isolated island.

These finches feed on seeds of different sizes from different species of plants. The finches show variation in the size of their beaks. Birds with larger beaks can eat large and small seeds. Birds with smaller beaks are only able to eat small seeds.

In 1977 there was a severe drought on the island. This killed many species of plants that the finches feed on. One species of food plant survived and produced large seeds.

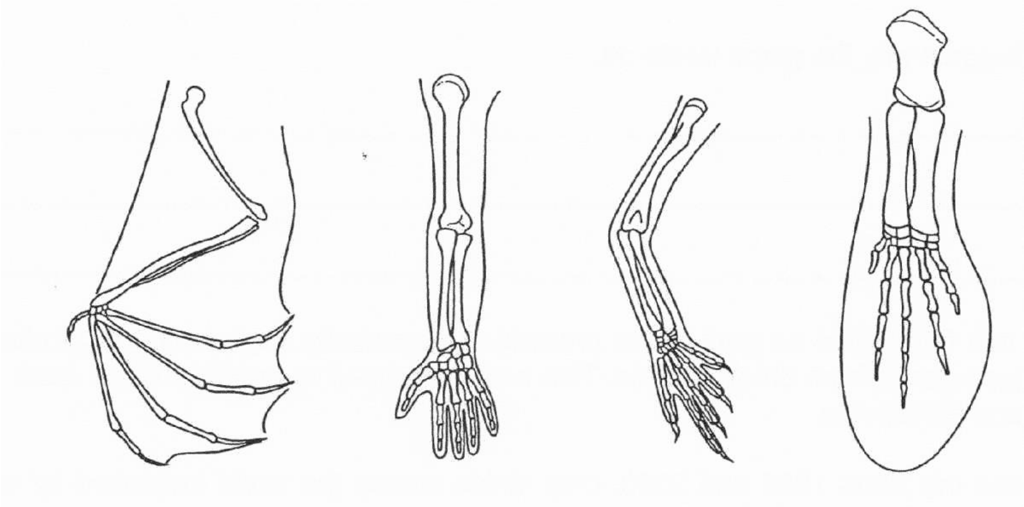
The graph shows the distribution of beak sizes of the finch population before the drought. Beak size was measured by the depth of the beak, as shown in the diagram.



Which row correctly shows the inheritance of beak size and the type of selection following drought?

	inheritance	type of selection
A	multiple alleles	directional
B	polygenic	directional
C	multiple alleles	stabilising
D	polygenic	stabilising

30. The diagram shows the arrangement of bones in the pentadactyl forelimbs of four vertebrates.



How does the arrangement of these bones support Darwin's theory of natural selection?

A They show how biotic factors operating on a variable genotype can bring about evolutionary change.

B They show how groups with separate evolutionary origins differ over time as a result of divergent evolution.

C They show how analogous structures have become adapted to perform different functions.

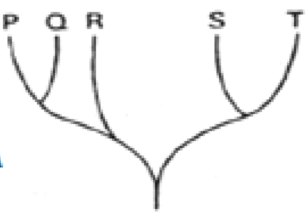
D They show how homologous structures have become adapted to perform different functions.

31. Five species, P, Q, R, S and T, possess the protein cytochrome *c* oxidase. The primary structure of the protein was determined for each species and the number of amino acid differences between species is given below.

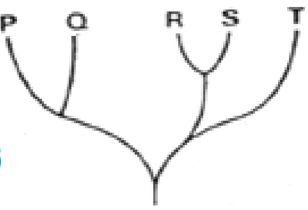
	P	Q	R	S	T
P	0	7	8	20	22
Q	7	0	3	19	17
R	8	3	0	18	21
S	20	19	18	0	10
T	22	17	21	10	0

From the evidence, which of the following represents the probable evolutionary relationship between the species?

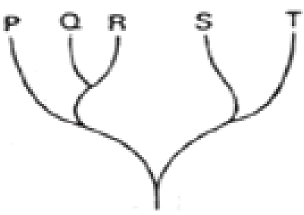
C



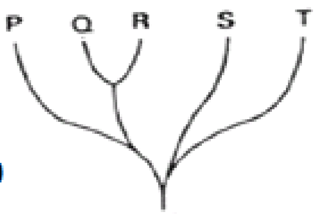
A



B



C



D

32. Which function is carried out by hormone glucagon?

A Glucagon catalyses the break down of glycogen to glucose in liver cells

B Glucagon activates enzymes involved in gluconeogenesis.

C Glucagon facilitates the entry of glucose into cell.

D Glucagon activates enzymes for glycogenesis.

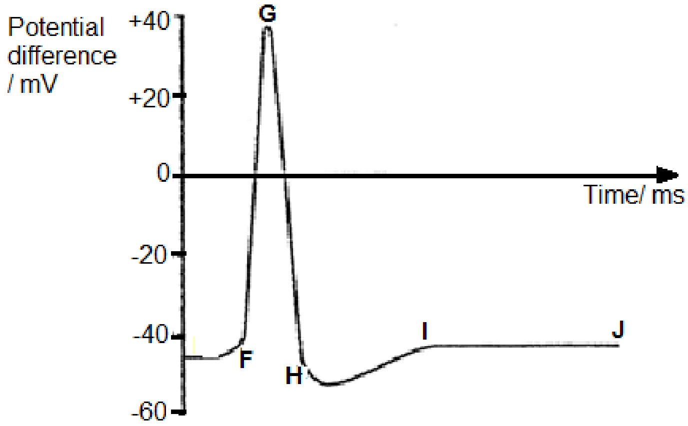
33. The table below shows a description of the activity of four drugs.

Drug	Description of activity
1	Enhance action of adenyl cyclase
2	Inhibit acetylcholinesterase
3	Block voltage gated Ca^{2+} channels
4	Block insulin receptors

Which of the following combination shows the consequence for each of the four drugs?

	Drug 1	Drug 2	Drug 3	Drug 4
A	Increase action potential initiation at pre-synaptic membrane	Decrease action potential initiation at pre-synaptic membrane	Block release of neurotransmitter into synaptic cleft	G-protein not activated
B	Activation of intracellular proteins without presence of signal	Decrease signal transduction efficiency	More neurotransmitter released into synaptic cleft	G-protein not activated
C	Activation of intracellular proteins without presence of signal	Increase action potential initiation at post-synaptic membrane	Block release of neurotransmitter into synaptic cleft	Insulin receptors do not dimerise
D	Decrease signal transduction efficiency	Increase action potential initiation at pre-synaptic membrane	More neurotransmitter released into synaptic cleft	Insulin receptors do not dimerise

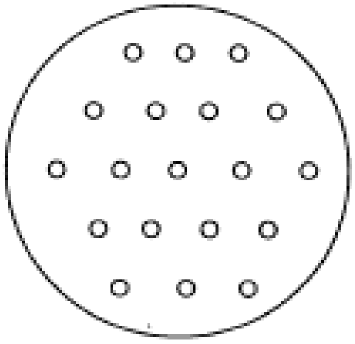
34. The graph shows the electrical events associated with a nerve impulse.



Which of the following correctly identifies the events at the respective phases?

	F-G	G-H	H-I	I-J
A	Opening of voltage-gated Na^+ channels	Voltage-gated Na^+ channel closes	Voltage gated K^+ channel closes	Sodium-potassium pump pumps out 3 Na^+ ions and takes in 2 K^+ ions
B	Opening of voltage-gated Na^+ channels	Voltage-gated Na^+ channel closes	Opening of voltage-gated Cl^- channels	Sodium-potassium pump pumps out 3 K^+ ions and takes in 2 Na^+ ions
C	Opening of voltage-gated K^+ channels	Voltage-gated Na^+ channel closes	Opening of voltage-gated Cl^- channels	Sodium-potassium pump pumps out 3 K^+ ions and takes in 2 Na^+ ions
D	Opening of voltage-gated K^+ channels	Opening of voltage-gated Cl^- channels	Voltage-gated Na^+ channel closes	Sodium-potassium pump pumps out 3 Na^+ ions and takes in 2 K^+ ions

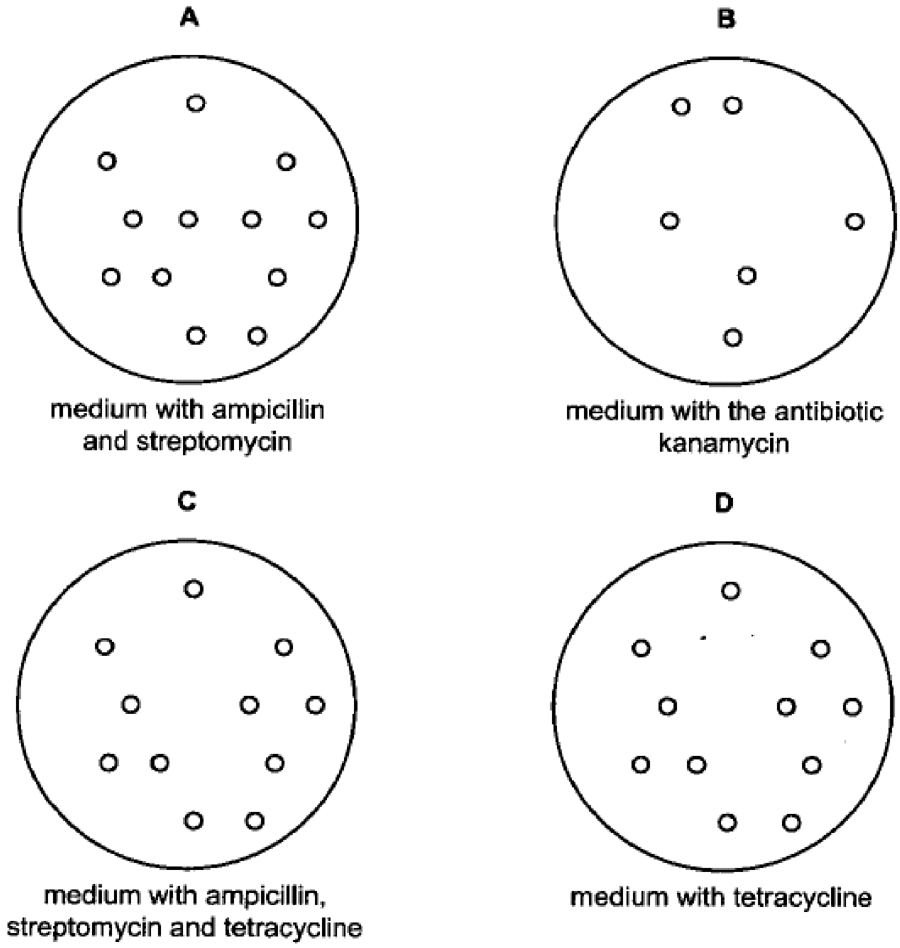
35. A gene coding for the production of a human gene product was inserted into a plasmid with genes coding for resistance to antibiotics ampicillin, streptomycin and tetracycline. The plasmids were used to transform *E. coli* and the bacteria grown on a nutrient medium. The resulting master plate is shown in the diagram.



Transformed cells were selected by replica plating the bacteria colonies onto media containing various antibiotics.

Which plate contains the colony of bacteria into which the human gene has been successfully inserted?

A



36. Which row correctly describes three steps of one cycle of a polymerase chain reaction?

	Step 1	Step 2	Step 3
A	DNA is denatured to single strands.	After cooling, primers bind to the 3' ends of the separated DNA strands	After heating, the DNA primers are elongated by DNA polymerase.
B	DNA is denatured to single strands	After further heating, primers bind to the 3' ends of the separated DNA strands	After cooling, the RNA primers are elongated by DNA polymerase.
C	DNA is denatured to single strands.	After cooling, primers bind to the 5' ends of the separated DNA strands.	After heating, the DNA primers are elongated by DNA polymerase.
D	DNA is denatured to single strands.	After further heating, primers bind to the 5' ends of the separated DNA strands	After cooling, the RNA primers are elongated by DNA polymerase.

37. There are two forms of the seeds of the garden pea, *Pisum sativum*, one with a smooth skin and the other with a wrinkled skin.

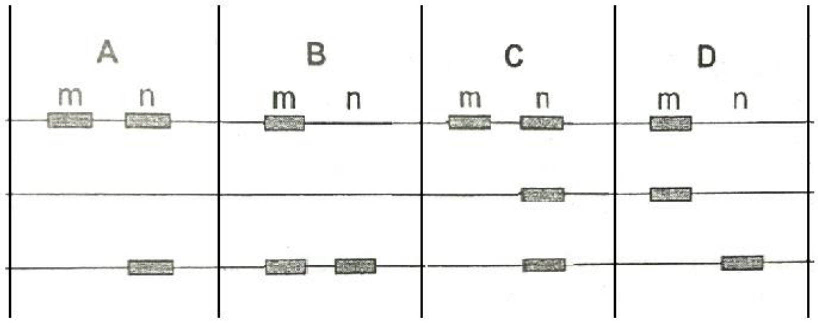
Plants with wrinkled seeds were known to have a recessive allele with an insertion of a transposable element into a gene determining the shape of the seed.

Two heterozygous pea plants were crossed. Two of the offspring (m and n) had the genomic DNA removed and cut by restriction enzyme at site flanking the gene. The DNA fragments were separated by electrophoresis.

The different alleles of the gene can be detected by a gene probe.

Which pattern of bands shows a wrinkled phenotype and a smooth phenotype?

positive plate (anode)



negative plate (cathode)

B

38. In parallel with the Human Genome Project, the DNA of a set of model organisms (such as bacteria, roundworms, fruitflies, mice and chimpanzees) are being studied.

Which technique might be most useful in using the information from model organisms to provide critical clues about the structure and function of human genes?

A Sequencing

B DNA fingerprinting

C Comparing molecular homology

D Finding the regions of repetitive sequences

39. What are normal functions of stem cells in a living human?

	Functions		
A	Differentiating into many kinds of cells in a 3-5 day old human embryo	Producing insulin in pancreas damaged by type 1 diabetes	Cells that can differentiate into bone cells in the skeleton
B	Differentiating into many kinds of cells in a 3-5 day old human embryo	Producing red blood cells worn out by normal wear and tear	Cells that can differentiate into cardiac muscle cells in the heart
C	Differentiating into only one kind of cell in a 3-5 day old human embryo	Producing dopamine in the brains of people with Parkinson's	Cells that can differentiate into cartilage cells in the joints
D	Differentiating into only one kind of cell in a 3-5 day old human embryo	Producing differentiated cells that can be used to screen new drugs	Cells that can differentiate into nerve cells in the brain

40. A gene for an insecticidal toxin was introduced into a crop plant using genetic modification. The toxin kills only a particular type of insect.

What is **not** likely to be affected by this genetic modification?

A ratio of population size between different insect species within the region

B growth of other crop plants within the region

C quantity of insecticide sprayed on the crop

D the number of insects resistant to the toxin

Paper 1

1.	A	11.	A	21.	B	31.	C
2.	D	12.	D	22.	D	32.	B
3.	C	13.	D	23.	D	33.	C
4.	B	14.	C	24.	C	34.	A
5.	D	15.	D	25.	C	35.	A
6.	D	16.	D	26.	B	36.	A
7.	B	17.	C	27.	D	37.	B
8.	C	18.	A	28.	A	38.	C
9.	C	19.	B	29.	B	39.	B
10.	A	20.	C	30.	D	40.	B